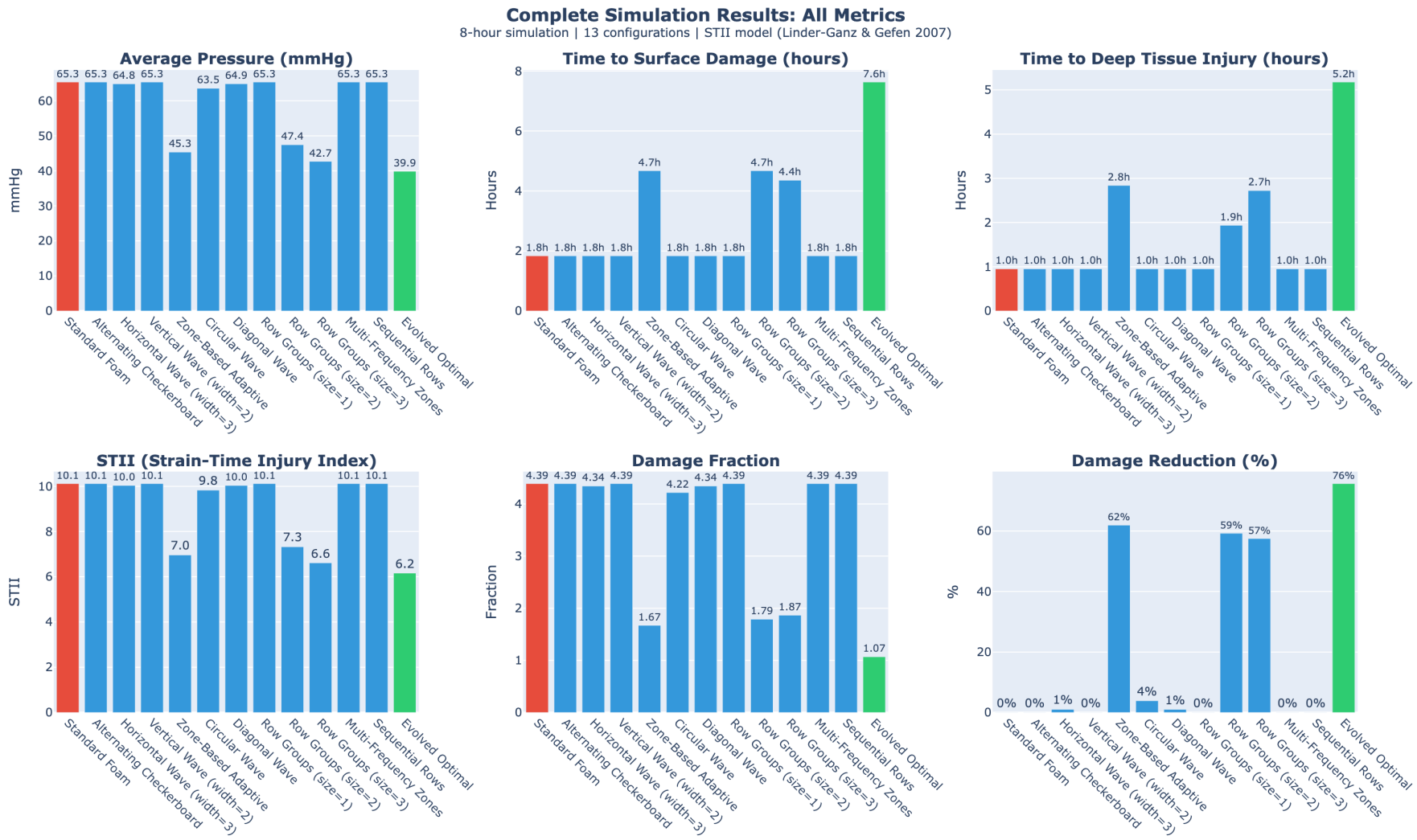
# Tissue Damage Analysis

8-hour simulation | STII model (Linder-Ganz & Gefen 2007)



## Table 1: Raw Simulation Data

| Configuration | Pressure | Surface (h) | DTI (h) | STII | Damage |
|---|---|---|---|---|---|
| Standard Foam | 65.34 | 1.8 | 1.0 | 10.12 | 4.39 |
| Alternating Checkerboard | 65.34 | 1.8 | 1.0 | 10.12 | 4.39 |
| Horizontal Wave (width=3) | 64.84 | 1.8 | 1.0 | 10.04 | 4.34 |
| Vertical Wave (width=2) | 65.34 | 1.8 | 1.0 | 10.12 | 4.39 |
| Zone-Based Adaptive | 45.34 | 4.7 | 2.8 | 6.96 | 1.67 |
| Circular Wave | 63.55 | 1.8 | 1.0 | 9.83 | 4.22 |
| Diagonal Wave | 64.86 | 1.8 | 1.0 | 10.03 | 4.34 |
| Row Groups (size=1) | 65.34 | 1.8 | 1.0 | 10.12 | 4.39 |
| Row Groups (size=2) | 47.44 | 4.7 | 1.9 | 7.33 | 1.79 |
| Row Groups (size=3) | 42.69 | 4.4 | 2.7 | 6.61 | 1.87 |
| Multi-Frequency Zones | 65.34 | 1.8 | 1.0 | 10.12 | 4.39 |
| Sequential Rows | 65.34 | 1.8 | 1.0 | 10.12 | 4.39 |
| Evolved Optimal | 39.87 | 7.6 | 5.2 | 6.16 | 1.07 |

## Table 2: Percent Change from Baseline

| Configuration | Pressure % | DTI % | STII % | Damage Red % |
|---|---|---|---|---|
| Standard Foam | Baseline | Baseline | Baseline | 0.0 |
| Alternating Checkerboard | +0.0 | +0.0 | +0.0 | 0.0 |
| Horizontal Wave (width=3) | -0.8 | +0.0 | -0.8 | 1.1 |
| Vertical Wave (width=2) | +0.0 | +0.0 | +0.0 | 0.0 |
| Zone-Based Adaptive | -30.6 | +198.2 | -31.2 | 61.9 |
| Circular Wave | -2.7 | +0.0 | -2.8 | 3.9 |
| Diagonal Wave | -0.7 | +0.0 | -0.8 | 1.0 |
| Row Groups (size=1) | +0.0 | +0.0 | +0.0 | 0.0 |
| Row Groups (size=2) | -27.4 | +103.5 | -27.6 | 59.2 |
| Row Groups (size=3) | -34.7 | +186.0 | -34.6 | 57.5 |
| Multi-Frequency Zones | +0.0 | +0.0 | +0.0 | 0.0 |

| | | | | |
|---|---|---|---|---|
| Sequential Rows | +0.0 | +0.0 | +0.0 | 0.0 |
| Evolved Optimal | -39.0 | +443.9 | -39.1 | 75.6 |

## Best Configuration: Evolved Optimal

| Metric | Value |
|---|---|
| Pressure Reduction | -39.0% |
| DTI Improvement | +443.9% |
| STII Reduction | -39.1% |
| Damage Reduction | 75.6% |
| DTI Efficiency | 544% |